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Studiengang / Fachsemester Study program / Subject semester

MBA 90 CP - Analytical skills for business

Prüfungsfach / Prüfungsdozent:in Examination subject / Examiner

1) Git is a distributed version Control System that is used manage code, track files and data also enable multiple users or contributors to work on the database at the same time.

The Core factors of Git include :

Local Repositories : a central storage location with revision of file history.

Commit: a snapshot of the project at a given time with changes.

Branch : a single line development, that enables multiple users to work parallelly.

Merge: enables to merge the changes made in different branches.

Github: is a web based cloud hosting platform for git repositories. It comes with collaborative and project management tools for developers.

Github allows features such as local repositories for independent users. It has staging area that is a intermediate zone to decide changes before committing to repository. Github allows limited branching & merging of data files.

Primary purpose: Git and Github are part of an ecosystem that enables a developer to commit changes and work on the database and "push" them to Github, so the files are accessed by the peers or people working on the same project. **Git**

b) The Core Concepts of Git are:

Repository: a central storage location where the files are saved in the database. A repository has track of the revision history and has audit trail.

Commit: It is a snapshot of a project at a particular time with the changes made.

Branch: a single line of development, where it allows multiple users to work on the file parallelly without changing the codebase.

⇒ In a collaborative analysis project core concepts of version control system ~~is~~ ^{are} needed to track back changes and revisions made.

⇒ Reproducibility and automation: automates workflows and repeated tasks for better productivity

- ⇒ Audit trail of changes made are available for accountability.
- ⇒ multiple users working on the same database can enhance collaborative work culture.
- ⇒ real time analytics and changes can be made to avoid further mistakes due to old or erroneous files.

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2) Python is a versatile programming language widely used in data analytics and programming. It comes with vast libraries such as "pandas", "matplotlib". Unlike other outdated language python enables great and better visualisations and integration with API's. This results in better statistical visualisation of graphs that can be used to communicate with non-technical stakeholders.

SQL: Structured Query language is a programming language used by data miners and strategical developers to undermine the large datasets to simpler form.

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In business analytics python & SQL play an important role; they come with features such as:

- i) Advanced visualisation of technical data that can be used in businesses for non-technical stakeholders.

- ii) Scalability issues : Unlike excel and outdated languages , Python & SQL can handle large dataset without much human intervention
- iii) Advanced integration of version Control System
- iv) Reproducibility and automation
- v) Personalisation & Customisation of datas based on customer needs enhances the prospects in future business.

b) The key elements in programming languages include :

Syntax : is a set of rules in the programming language that allows the code to function in the right manner.

Libraries : a set of pre determined codes that can be used by users to save time in data framing & projects.

Variables : is a name in the given file or folder. A variable enables to track the particular code or identify them.

Loops : a loop is a continuous form of objects that are repeating in the trail of codes.

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all the factors are important in data analysis to form a structured form of base of code that aligns to basic principles and are important for error free results. 9/9

- a) No code - Low code
- b) Advantages of no code - Low code tools are:
- i) Advanced Scalability : large datasets can be cumbersome to work without errors, but low code tools enable to crack them with less human intervention.
 - ii) Reproducible and automated workflow: they offers audit trail backs to revisions made and reproduce repetitive tasks. this makes the data engineer to focus on better tasks.
 - iii) Advanced Visualisations: better graphical representations using libraries such as 'matplotlib', pandas.
 - iv) Better integration with API and other data interface almost offers seamless integration.
 - v) Version Control integration: most platforms loss data due to this update. low code tools enable version control integration without data loss.

vi) Customisation and personalisations of the codes based on customer preference enhances the relationship & Customer satisfaction.

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4) Central tendency:

i) Mean: It is the arithmetic average of all the given values. It is sensitive to outliers.

ii) Median: It is the middle value of the ordered data. which is robust to outliers.

iii) Mode: is the strategic value or the 50% of the given data.

⇒ The most appropriate measure of central tendency would be median or IQR (Interquartile range) as they are not extremely affected by the outliers.

⇒ Median can be used when the numbers or values in the analytics have extreme differences.

⇒ Mean can be used to determine the average when the numerical data falls within the same range of numbers. This can be used to determine the middle ground of the data.

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b) Univariate data is the analysis of only one variable in data analysis. Example: the sales expenditure of a company is determined using bar plots for the entire year.

Bivariate data analysis uses two variables from the given data to determine statistics.

eg: the monthly expenditure vs the sales made are compared for the year. histograms, bar plots, are used.

Multivariate data involves 2 or more variables to determine the data analysis.

eg: customer satisfaction, retention rate, sales made, expenditure of a company are taken to determine the stats graphically using scatter plots or box plots.

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c) Missing values: One major pitfall in this category is misinterpreting data. Most analysts often read the data without understanding the underlying concepts resulting in human error & wrong statistical decisions made.

Inconsistent Data: Another pitfall here includes lack of understanding after reading the data. Here graphs and other data related decisions are slightly distorted affecting the results.

- iii) Duplicate records: Another pitfall that can be done while creating data or modifying existing data. Data manipulation or fabrication can be resulted due to unknown factors or hidden agendas made by the analysts.

Appropriate strategies include: Data analysts to not cherry pick successful case studies, as this results in biases and skewed analysis. Each aspect of the cases to be determined to avoid human error. This includes both successful and unsuccessful cases.

5) Practical R programming

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a) $3 + \sqrt{9} - 2^2 \cdot \frac{4}{3}$

$\Rightarrow \text{result} \leftarrow 3 + \text{sqrt}(9) - 2^{\{2\}} * 4/3$

b) `getwd()`

c) `?mean()`

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d) `sales_data |>`

`Region <- c("North", "South", "East")`

`Sales <- c("45000", "38000", "52000")`

2) d) Popular IDEs used are:

i) R studio

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